

Bell Nozzle Rocket Engine Equation Database

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$$C_nH_{2n+2} + \frac{3n+1}{2}O_2 \Longrightarrow (n+1)H_2O + nCO_2 + energy \quad (1)$$

$$\dot{m} = \frac{m}{t} \quad (2)$$

$$\frac{1}{\gamma-1} = \sum \frac{mol\%}{\gamma_i-1} \quad (3)$$

$$P_s = P_x \left(1 + \frac{\gamma-1}{2} Ma_x^2 \right)^{\frac{\gamma}{\gamma-1}} \quad (4)$$

$$T_s = T_x \left(1 + \frac{\gamma-1}{2} Ma_x^2 \right) \quad (5)$$

$$V_e = \sqrt{\frac{2\gamma}{\gamma-1}RT_c \left(1 - \left(\frac{P_e}{P_c} \right)^{\frac{\gamma-1}{\gamma}} \right)} \quad (6)$$

[1]

References

[1] test, "test." test, test.